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| **Fiscal Year:** | FY 2022 | **Task Last Updated:** | FY 11/30/2021 |
| **PI Name:** | Dannemiller, Karen Ph.D. | | |
| **Project Title:** | Humidity and Microbial Growth in ISS Dust (HUMID) | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Microbiology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential<br>(2) Bioregenerative Life Support | | |
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| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research |
| **Start Date:** | 01/31/2019 | **End Date:** | 01/31/2023 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA ARC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 1/31/2023 per F. Hernandez/ARC (Ed., 6/20/22). | | |
| **Key Personnel Changes/Previous PI:** | Nick Nastasi graduated with his MS and became a PhD student working on the project. Ashleigh Bope defended her dissertation and will graduate December 2021. | | |
| **COI Name (Institution):** | Horack, John Ph.D. ( Ohio State University )<br>Meyer, Marit Ph.D. ( NASA Glenn Research Center ) | | |
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| **Task Description:** | In space, astronauts are confined in spacecraft with recirculated air and altered microbial exposures. These exposures have important implications for astronaut health (allergens, pathogens) [1], spacecraft integrity (prevention of microbial-induced corrosion such as on Mir [2]), and planetary protection from microbes. Microbial communities on the International Space Station (ISS) have been characterized [3], [4], and this project would expand upon that work with novel chamber experiments to allow prediction of microbial communities and functions under different conditions. Moisture is a critical factor in determining the presence and growth of microorganisms. In fact, our recent work demonstrated that microbial growth in house dust can make a substantial contribution to human exposure under elevated relative humidity (RH) conditions (>80% RH) [5]. Moisture is inherent to an occupied spacecraft environment because water is necessary for life. Astronauts sweat, bathe, and exhale moisture, and the resulting water vapor and condensation occurs on the ISS and will occur on all future crewed missions. In fact, Mir was damaged by fungi that could degrade metals, glass, or plastics prior to its decommissioning [2]. Microbial growth may occur directly on surfaces, but also likely occurs in the dust. The dust represents a rich source of organic material and water is most likely the main factor limiting growth. Furthermore, dust is an important source of human exposure and could potentially contaminate other planets.<br><br>However, microbial growth on spacecraft is only beginning to be characterized [6], and we do not know which microbes will grow in the dust itself. Areas of high moisture could potentially harbor allergens, pathogens, and/or microbes capable of material degradation and planetary contamination. Currently, we have characterized neither a healthy nor unhealthy spacecraft microbiome.<br><br>This project will utilize vacuumed dust to be returned from the ISS in 2018 from collection in "Divert Unwanted Space Trash (DUST)." The goal of this project is to characterize microbial communities and function that may occur on the ISS in areas of high relative humidity and condensation.<br><br>Aim 1: Characterize microbial growth in ISS dust under different relative humidity conditions in a laboratory chamber study. Dust from the ISS will be incubated under varying relative humidity conditions [5]. We will characterize microbial growth using DNA-based methods to quantify growth and determine microbial communities present in both the original dust and incubated samples.<br><br>Aim 2: Identify potential microbial function that can occur in dust on board the ISS. We will extract RNA from a subset of samples for metatranscriptomic analysis of fungal communities. Our yet unpublished work demonstrates that microbial function occurs even at 50% RH in house dust.<br><br>Our results will go beyond microbial measurement to predict moisture-related changes to the indoor microbiome and astronaut exposure. Our work will provide added value to microbial data collected previously on the ISS. For instance, Microbial Tracking 1 ( https:// ) characterized airborne and surface-associated populations of microorganisms aboard the ISS. The "Humidity and Microbial Growth in ISS Dust (HUMID)" investigation complements this data through use of a chamber study observing fundamental microbial processes that may occur under conditions of increased moisture in occupied spacecraft. We need to understand how to characterize the spacecraft microbiome as either healthy or unhealthy. Ultimately, we will be able to develop improved microbial clearance protocols that are more relevant to astronaut health, planetary protection, and spacecraft integrity. Our design decisions in space have important implications in controlling the microbes that are present. By understanding the fundamental processes that occur in this environment, we can design for a healthy microbiome instead of being subject to unintended consequences.<br><br>References<br><br>[1] M. J. Mendell, A. G. Mirer, K. Cheung, M. Tong, and J. Douwes, “Respiratory and Allergic Health Effects of Dampness, Mold, and Dampness-Related Agents: A Review of the Epidemiologic Evidence,” Environ. Health Perspect., vol. 119, no. 6, pp. 748–756, 2011.<br><br>[2] N. Novikova, P. De Boever, S. Poddubko, E. Deshevaya, N. Polikarpov, N. Rakova, I. Coninx, and M. Mergeay, “Survey of environmental biocontamination on board the International Space Station,” Res. Microbiol., vol. 157, no. 1, pp. 5–12, Jan. 2006.<br><br>[3] N. K. Singh, A. Blachowicz, A. Checinska, C. Wang, and K. Venkateswaran, “Draft Genome Sequences of Two Aspergillus fumigatus Strains, Isolated from the International Space Station.,” Genome Announc., vol. 4, no. 4, Jul. 2016.<br><br>[4] A. Checinska, A. J.Probst, P. Vaishampayan, J. R. White, D. Kumar, V. G. Stepanov, G. E. Fox, H. R. Nilsson, D. L. Pierson, J. Perry, and K. Venkateswaran, “Microbiomes of the dust particles collected from the International Space Station and Spacecraft Assembly Facilities,”Microbiome, vol. 3, no. 1, p. 50, Dec. 2015.<br><br>[5] K. C. Dannemiller, C. J. Weschler, and J. Peccia, “Fungal and bacterial growth in floor dust at elevated relative humidity levels,” Indoor Air, no. May, pp. 1–10, 2016.<br><br>[6] C. M. Ott, R. J. Bruce, and D. L.Pierson, “Microbial Characterization of Free Floating Condensate Aboard the Mir Space Station,” Microb. Ecol., vol. 47, no. 2, pp. 133–136, 2004. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This work broadens understanding of microbial growth in dust, which is relevant both on the International Space Station (ISS) and in many environments on Earth. This includes homes, offices, schools, other buildings, and specialized environments such as aircraft and submarines. Our work provides a framework with which to model this growth and allow potential prediction of conditions that could lead to adverse outcomes. The microbial growth in dust has implications both for material corrosion and human health. |

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| **Task Progress:** | The goal of this project is to characterize microbial communities and microbial function that could possibly occur on the International Space Station (ISS) or other spacecraft in areas of high relative humidity (RH) and condensation. This will mimic an unanticipated high-moisture event and allow for prediction of microbial contamination to inform remediation strategies. We will achieve this goal through the following aims:<br>Aim 1: Characterize microbial growth in ISS dust under different relative humidity conditions in a laboratory chamber study. Dust from the ISS has been returned to Earth following an existing collection experiment called “Divert Unwanted Space Trash (DUST).” During their regularly scheduled cleaning routine, crewmembers remove the debris from vents and HEPA filter screens throughout the ISS and it is collected in vacuum bags. This dust will be incubated under constant relative humidity conditions (50%, 60%, 70%, 80%, 85%, 90%, and 100% RH) for 2 weeks using established protocols. In addition, the time-of-wetness (TOW) framework will be used to model microbial growth under daily variations in RH for elevated (85% RH) and saturated (100% RH) conditions 0, 6, 12, 18, and 24 hours per day from 50% RH. We will characterize microbial growth using quantitative Polymerase Chain Reaction (qPCR) and high-throughput DNA sequencing to quantify growth and determine microbial communities present in both the original dust and incubated samples.<br><br>Aim 2: Identify potential microbial function that can occur in dust on board the ISS. We will incubate a subset of samples at 50%, 85%, and 100% RH and extract RNA for metatranscriptomic analysis of fungal communities. We will look for metabolic activity that occurs at elevated moisture conditions. These include secondary metabolic processes, allergen and toxin production activities, and parthenogenesis activities. We will also examine expression patterns among sample replicates to explore how moisture and time of sample collection impact gene expression. Our previous work demonstrates that microbial function occurs even at 50% RH in house dust.<br><br>At this stage in the project, we have received dust from four different vacuum bags that were returned from the ISS HEPA filter coverings, which are part of the space station's ventilation system. These bags were returned to Earth for analysis over a period of 12 months in 2019 and 2020. We have completed many of the Aim 1 experiments, with a small number remaining. Dust incubations for all ISS vacuum bags have been completed. Total fungi and bacteria have been quantified with qPCR for all incubations using universal fungal and bacterial primers. This includes all 2-week incubations and time-of-wetness incubations. The time-of-wetness analysis revealed that this framework can be used for prediction of fungal contamination following an unexpected high-moisture event. Visual analyses of microbial growth in the ISS dust have also been performed via scanning electron microscopy (SEM), revealing fungal structures indicative of reproduction such as spores, hyphae, and fruiting bodies. Growth patterns on the dust were similar to fungal growth patterns on dust and carpet from buildings on Earth.<br><br>We have completed DNA sequencing and are currently analyzing the results to understand changes in microbial community composition. Additional chemical analyses were performed on the ISS dust at elevated RH conditions, including total organic carbon (TOC) and other potentially relevant microbial nutrients (S, P, Na, Ca, Mg, K). We have completed all experiments for Aim 2 and have processed most of the data. Samples from all four vacuum bags have been incubated in triplicate at 50%, 85% and 100% RH and RNA has been extracted. Samples were reverse transcribed into cDNA and sequenced using Illumina NovaSeq. A polyadenylated RNA (polyA) selection protocol was used to select for eukaryotic RNA only. De novo metatranscriptome assembly was carried out using Trinity software and its accompanying packages. The alignment-based quantification method, RNA-Seq by Expectation-Maximization (RSEM), was used to estimate transcript abundance in each sample. Analysis of differentially expressed transcripts and clustering of transcripts according to expression profiles was performed leveraging DESeq2. Pairwise comparisons among each RH condition were performed. Contigs were annotated using Trinotate, a comprehensive annotation software suite designed for automatic functional annotation of de novo assembled transcriptomes. KEGG orthology (KO) terms and taxonomic identifications were determined for each contig using KEGG’s GhostKOALA (KEGG Orthology And Links Annotation) tool.<br><br>Results indicate that there are significant differences in gene expression as a function of moisture condition. Differences in gene expression between the different bags were not as important as the incubated RH condition. We are currently evaluating potential implications for human health and spacecraft integrity.<br><br>This work has been shared at several international/national meetings to diverse audiences.<br><br>Future plans: A detailed analysis of sequencing data will be conducted to see how microbial communities and function change as relative humidity changes. We will continue data analysis and manuscript writing. |
| **Bibliography Type:** | Description: (Last Updated: 12/08/2025) |
| **Abstracts for Journals and Proceedings** | Nastasi N, Bope A, Meyer M, Horack J, Dannemiller KC. "The effects of relative humidity on fungal growth in dust collected from the International Space Station (ISS)." 38th Annual Conference of the American Association for Aerosol Research, Virtual Meeting, October 5-9, 2020.<br>Abstracts. 38th Annual Conference of the American Association for Aerosol Research, Virtual Conference, October 5-9, 2020:A412. , Oct-2020 |
| **Articles in Other Journals or Periodicals** | Nastasi N, Renninger N, Bope A, Cochran SJ, Greaves J, Haines SR, Balasubrahmaniam N, Stuart K, Panescu J, Bibby K, Hull NM, Dannemiller KC. "Viability of MS2 and Phi6 Bacteriophages on carpet and dust." bioRxiv preprint server. Posted May 18, 2021. https://doi.org/10.1101/2021.05.17.444479 , May-2021 |
| **Articles in Other Journals or Periodicals** | Nastasi NA, Bope A, Dannemiller KC, Meyer ME, Horack JM. "Dust from astronauts and space habitats (DASH): Implications on long term space missions." National Academies of Sciences, Engineering, and Medicine. To appear in special decadal survey on biological and physical sciences research in space 2023-2032. In review. , Jun-2023 |